

α -labeling of banana tree graphs

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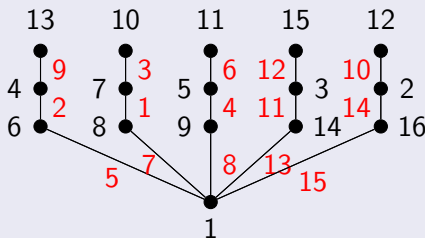
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Graceful labeling of a tree

Definition [A. Rosa, 1967.]

We say that a labeling of a graph G is *graceful* if $\text{Im } f \subseteq [m+1]$, the induced edge labeling $f^* : E \rightarrow \mathbb{N}$ defined by $f^*(vw) = |f(v) - f(w)|$ is injective and $\text{Im } f^* = \{f^*(vw) : vw \in E\} = [n-1]$.

Example



Conjecture [Ringel-Kotzig]

All trees are graceful.

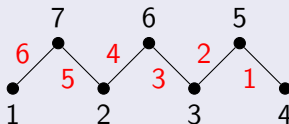
α -labeling of a tree

Definition [A. Rosa, 1967.]

A labeling of a graph G is an α -labeling if it is graceful and if there exists a constant $\alpha \in \mathbb{N}$ such that

$$f(v) \leq \alpha < f(w) \text{ or } f(w) \leq \alpha < f(v), \text{ for all } vw \in E.$$

Example



α -labeling of a tree with $\alpha = 4$

- Applications of α -labelings: coding theory, graph decompositions

Conjugate and reverse α -labelings

Remark [A. Rosa, 1967.]

- 1 If f is an α -labeling of G and $f^*(vw) = 1$ for some $v, w \in V$, then $\alpha = \min\{f(v), f(w)\}$.
- 2 If f is an α -labeling of a graph G with n vertices, then its complementary labeling $\bar{f}$ defined as $\bar{f}(v) = n + 1 - f(v)$ is also an α -labeling of G with $\bar{\alpha} = n - \alpha + 1$.
- 3 If f is an α -labeling of G , its reverse (or inverse) labeling f^r defined as

$$f^r(v) = \begin{cases} \alpha + 1 - f(v), & f(v) \leq \alpha \\ n + \alpha + 1 - f(v), & f(v) > \alpha \end{cases}$$

is also an α -labeling of G with $\alpha^r = \alpha$.

Definition of a banana tree $B(n, k)$

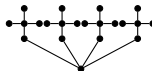
- Focus of our work: α -**labeling** and a special family of trees – banana trees
- Hierarchical structure: n star graphs S_k
- Each star is connected to a central vertex (root)



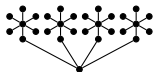
$B(2, 4)$



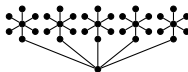
$B(3, 4)$



$B(4, 5)$



$B(4, 7)$



$B(5, 7)$

Rooted Standard Form and isomorphic labelings

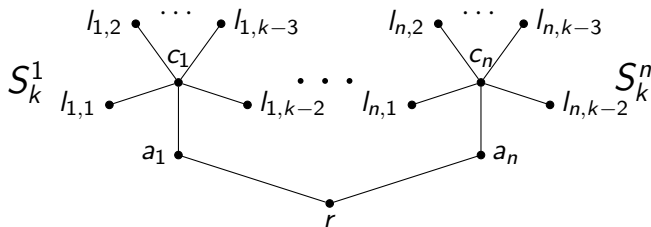


Figure: Rooted standard form of a $B(n, k)$ tree

We introduced notation for simpler representation of α -labelings of $B(n, k)$ trees:

- The set of all leaves of the i -th star – $\mathcal{L}_i = \{l_{i,j} \mid j \in [k-2]\}$
- The set of all non-central nodes of the i -th star – $\mathcal{B}_i = \{a_i\} \cup \mathcal{L}_i$

Note

Swapping the labels of two leaves or two stars doesn't change the labeling.

Theorem [G. Sethuraman; J. J. Jesintha, 2009.]

All banana trees are graceful.

Theorem

- 1 All complete bipartite graphs $K_{m,n}$ are α -labelable. ($B(n, 1) \cong K_{1,n}$) [A. Rosa, 1967.]
- 2 The banana tree $B(n, k)$ is α -labelable if $k > n$. [S. Nikson; P. S. Ikhlas; A. S. Kiki, 2023.]

The set of labelings $B(n, k)$

Definition [BG, Nakic, 202?]

Let $\mathcal{A}(n, k)$ be the set of all α -labelings of the $B(n, k)$ tree and $\mathcal{A}_x(n, k)$ be the set of all α -labelings of the $B(n, k)$ tree where $f(r) = x$. Let the set of principal α -labelings $\mathcal{P}(n, k)$ be defined as $\mathcal{P}(n, k) = \bigcup_{x=1}^{\lfloor \frac{n}{2} \rfloor + 1} \mathcal{A}_x(n, k)$. Lastly, let the set of *central* α -labelings $\mathcal{C}(n, k)$ be defined as $\mathcal{C}(n, k) = \mathcal{A}_{\lfloor \frac{n}{2} \rfloor + 1}(n, k)$.

Proposition [BG, Nakic, 202?]

Let f be an α -labeling of the $B(n, k)$ tree for any $n \in \mathbb{N}$ and $k \geq 2$. Then,

$$f(r) \leq \alpha \iff f(c_i) \leq \alpha \quad \forall i \in [n]$$

Partitioning $\mathcal{A}(n, k)$

Theorem [BG, Nakic, 202?]

For every $n, k \in \mathbb{N}$, we have

$$\mathcal{A}(n, k) = \mathcal{P}(n, k) \cup \mathcal{P}^r(n, k) \cup \overline{\mathcal{P}}(n, k) \cup \overline{\mathcal{P}^r}(n, k),$$

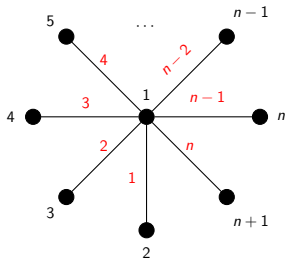
where $\mathcal{P}^r(n, k) = \{f^r : f \in \mathcal{P}(n, k)\}$, $\overline{\mathcal{P}}(n, k) = \{\bar{f} : f \in \mathcal{P}(n, k)\}$ and $\overline{\mathcal{P}^r}(n, k) = \{\bar{f}^r : f \in \mathcal{P}(n, k)\}$.

- Reducing the set of possible α -labelings by a factor of 4
- Reducing the complexity of the proofs

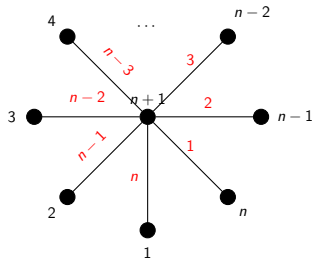
Classifying all the α -labelings of all the $B(n, 1)$ trees

Theorem [BG, Nakic, 202?]

The $B(n, 1)$ tree is α -labelable for all $n \in \mathbb{N}$. For $n = 1$ there exists a unique α -labeling. For $n \geq 2$ there exist exactly 2 distinct α -labelings.



α -labeling f_1 with $\alpha = 1$



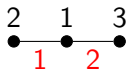
α -labeling f_2 with $\alpha = n$

Figure: The two α -labelings of $B(n, 1)$

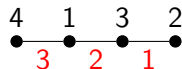
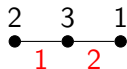
Classifying all the α -labelings of all the $B(1, k)$ trees

Theorem [BG, Nakic, 202?]

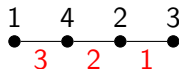
The $B(1, 2)$ tree and the $B(1, 3)$ tree have exactly 2 distinct α -labelings.



The α -labelings of $B(1, 2)$



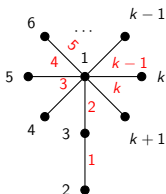
The α -labelings of $B(1, 3)$



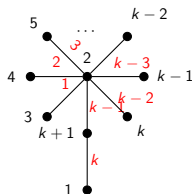
Theorem [BG, Nakic, 202?]

The $B(1, k)$ tree is α -labelable for $k \geq 4$ and has exactly 4 distinct α -labelings.

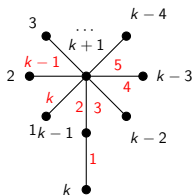
Classifying all the α -labelings of all the $B(1, k)$ trees



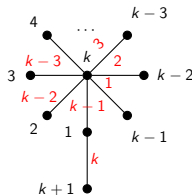
α -labeling f_1 with $\alpha_1 = 2$



α -labeling f_2 with $\alpha_2 = 2$



α -labeling f_3 with $\alpha_3 = k - 1$



α -labeling f_4 with $\alpha_4 = k - 1$

Classifying all the α -labelings of all the $B(2, k)$ trees

Theorem [BG, Nakic, 202?]

$\mathcal{A}_1(2, 2) = \emptyset$ and for every $k \geq 3$, we have $\mathcal{A}_1(2, k) = \{f\}$, where

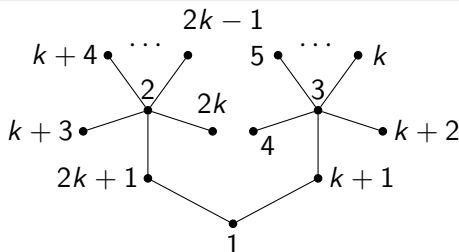
$$f(r) = 1$$

$$f(c_i) = i + 1$$

$$f(a_i) = (3 - i)k + 1$$

$$\mathcal{F}(\mathcal{L}_1) = \{k + 3, k + 4, \dots, 2k\}$$

$$\mathcal{F}(\mathcal{L}_2) = \{4, 5, \dots, k - 1, k, k + 2\}$$



α -labeling $f \in \mathcal{A}_1(2, k)$ with $\alpha = 3$

Classifying all the α -labelings of all the $B(2, k)$ trees

Theorem [BG, Nakic, 202?]

For every $k \geq 2$, we have $\mathcal{A}_2(2, k) = \{f_y \mid k + 3 \leq y \leq 2k + 1\}$, where

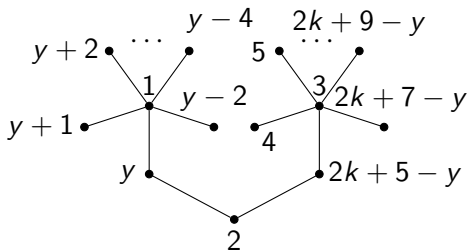
$$f_y(r) = 2, \quad f_y(c_1) = 1, \quad f_y(c_2) = 3$$

$$f_y(a_1) = y$$

$$f_y(a_2) = 2k + 5 - y$$

$$\mathcal{F}_y(\mathcal{L}_1) = \{y + 1, y + 2, \dots, 2k + 1\} \cup \{y - 2i \mid 1 \leq i \leq y - k - 3\}$$

$$\mathcal{F}_y(\mathcal{L}_2) = (\{4, 5, \dots, 2k + 4 - y\} \cup \{2k + 5 - y + 2i \mid 1 \leq i \leq y - k - 3\})$$



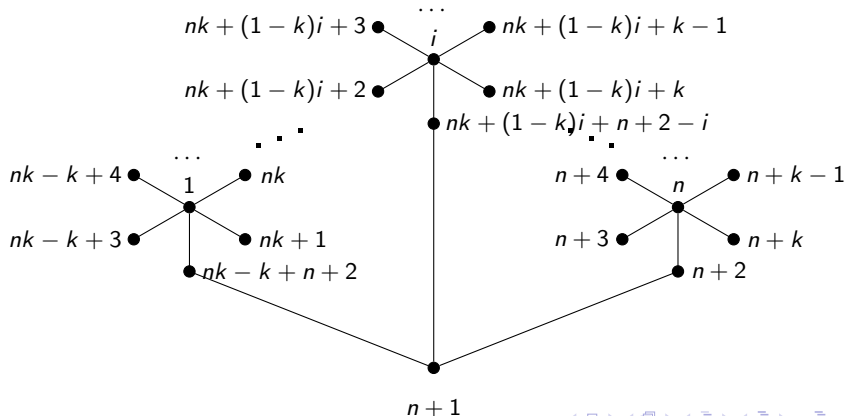
α -labeling $f_y \in \mathcal{A}_2(2, k)$ with $\alpha = 3$

A sketch of the proof for $B(n, k)$ for $k > n$

Theorem [BG, Nakic, 202?]

The $B(n, k)$ tree is α -labelable for all $n \geq 3$ and $k > n$.

Labeling which proves the theorem:



A sketch of the proof for $B(n, k)$ for $\frac{n}{2} + 1 \leq k \leq n$

Theorem [BG, Nakic, 202?]

The $B(n, k)$ tree is α -labelable for all $n \in \mathbb{N}$ and $\frac{n}{2} + 1 \leq k \leq n + 1$.

Labeling which proves the theorem:

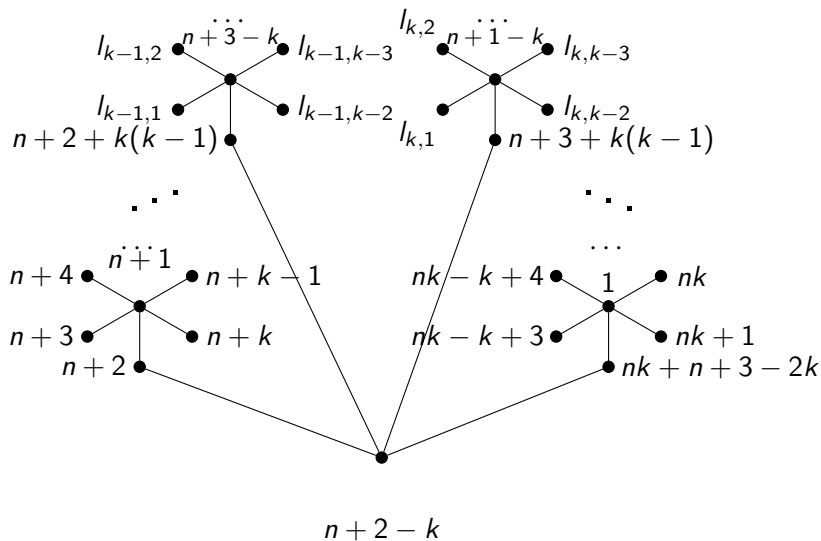
$$f(r) = n + 2 - k$$

$$f(c_i) = \begin{cases} n + 2 - i, & i < k \\ n + 1 - i, & i \geq k \end{cases}$$

$$f(a_i) = \begin{cases} n + 2 - k + ki, & i < k \\ n + 3 - 2k + ki, & i \geq k \end{cases}$$

$$\mathcal{F}(\mathcal{L}_i) = \{n + 3 - k + (k - 1)i, \dots, n + 1 + (k - 1)i\} \setminus \{f(a_i)\}$$

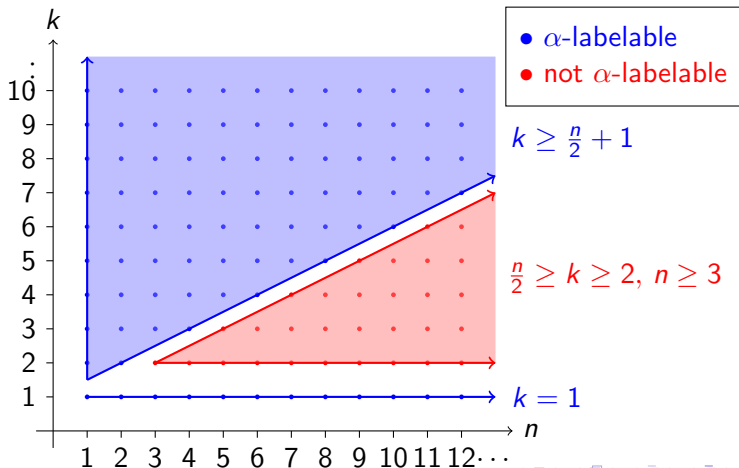
A sketch of the proof for $B(n, k)$ for $\frac{n}{2} + 1 \leq k \leq n$



α -labelability of $B(n, k)$ trees

Corollary [BG, Nakic, 202?]

The $B(n, k)$ tree is α -labelable if and only if either $k \geq \frac{n}{2} + 1$ or $k = 1$.



Conclusion

- We introduced new notation and concepts for α -labelings of $B(n, k)$ trees
- We **solved** the question of α -labelability of $B(n, k)$ trees for all values of n and k
- **Open question:** What is the number of non-isomorphic α -labelings of $B(n, k)$ trees for $n \geq 3$?

Thank you for your attention!

Any questions?